Training Day 9

1. Topics Covered and Brief Explanation

1. Introduction to Data Preprocessing

Data preprocessing is the process of preparing raw data into a clean and organized format before using it for Machine Learning. It helps improve data quality and the accuracy of model predictions.

2. Handling Missing Values

Missing values occur when some information is unavailable in a dataset. Studied different ways to handle missing data, such as:

- Removing rows containing missing values.
- Replacing missing values with the mean or median.
- Filling missing values with a suitable default value.

3. Removing Duplicate Records

Duplicate records are repeated entries in a dataset. Learned how to identify and remove duplicate rows to avoid unnecessary repetition and improve data quality.

4. Handling Inconsistent Data

Studied how inconsistent data, such as different spellings, unnecessary spaces and incorrect formats, can affect data analysis. Learned basic techniques to correct these inconsistencies.

5. Introduction to Data Normalization

Learned the basic concept of normalization, which involves scaling numerical values into a common range so that features with different scales can be handled more effectively by machine learning algorithms.

2. Tools and Technologies Used

Tool	Purpose
Python	Performing basic data cleaning operations.
Pandas	Handling missing values and duplicate records.
NumPy	Working with numerical values and missing data.
Google Colab	Writing and executing data preprocessing programs.

3. Practical Learning / Code Examples

Example 1: Handling Missing Values

```
import pandas as pd

data = {
```

```
    "Name": ["Aman", "Riya", "Karan"],
    "Marks": [80, None, 90]
}
df = pd.DataFrame(data)
df["Marks"] = df["Marks"].fillna(
    df["Marks"].mean()
)
print(df)
```

Explanation:

- None represents a missing value.
- isnull() can be used to identify missing values.
- fillna() replaces missing values.
- mean() calculates the average marks.

Example 2: Removing Duplicate Records

```
import pandas as pd
data = {
    "Name": ["Aman", "Riya", "Aman"],
    "Marks": [80, 90, 80]
}
df = pd.DataFrame(data)
df = df.drop_duplicates()
print(df)
```

Explanation:

The drop_duplicates() function removes repeated rows from the dataset.

4. Learning Outcomes

- Understood the importance of data preprocessing in Machine Learning.
- Learned different methods of handling missing values.
- Gained knowledge of identifying and removing duplicate records.
- Understood how to handle inconsistent data.
- Practised basic data cleaning operations using Pandas and Python.